

AYESHA NUSRAT

Software Engineer

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OBJECTIVE

Highly motivated and passionate about contributing to a well-established and growth-oriented organization where innovative ideas can be translated into impactful actions. Committed to continuous learning, professional excellence, and delivering meaningful results.

TECHNICAL SKILLS

- **Languages:** C++, C#, Python, SQL (T-SQL), TypeScript, JavaScript (ES6+), HTML5, CSS3.
- **Frameworks & UI Libraries:** .NET Core / .NET 5+, ASP.NET Web API, Blazor, Angular (18), Entity Framework Core, SignalR, Bootstrap.
- **Architecture & Communication:** Microservices, Monolithic to Microservices Evolution, GraphQL, RESTful APIs.
- **Databases & Performance:** MS SQL Server, NoSQL, Vector DB, SQL Profiler, Database Index Tuning, Query Optimization.
- **DevOps & Tooling:** CI/CD (Azure DevOps / GitHub Actions), Docker, Git, Azure / AWS.

WORK HISTORY

Senior Software Engineer, TKXEL

DEC 2025 - Present

Project: ASAP Semiconductor Ecosystem

- **Performance Optimization:** Designed and implemented a global Content Delivery Network (CDN) strategy for static content, reducing server load by 30% and significantly improving page-load times for international users.
- **Cost Efficiency:** Reduced cloud egress/data-transfer costs by optimizing how directly the server responds to requests.
- **Technical SEO Integration:** Partnered with the SEO team to implement advanced technical SEO, including Schema markup, Core Web Vitals improvements, and Server-Side Rendering (SSR), substantially increasing organic search visibility.
- **Scalable Web Engineering:** Led development of high-traffic modules of the ASAP Semiconductor e-commerce platform, architected to manage millions of SKUs and generate real-time quotes.

Project: FlowGenius – AI-Assisted Accounts Payable Platform

Technologies: .NET / Blazor (Razor), C#, Entity Framework Core, SQL Server, Hangfire, AI Invoice Extraction

- **End-to-End AP Workflow:** Built a demonstrable invoice-to-payment platform in a 10-day hackathon sprint (team of 7), covering AI-extracted invoice data, role-based access across 5 roles (Requestor, AP, HOD, Verifier, SuperAdmin), and dual-control payment verification.
- **Payment State Machine:** Implemented a multi-stage approval workflow (Draft → Submitted → AP → HOD Approval → Payment Marked → Verified) across 13 .NET controllers and Razor screens, backed by Entity Framework Core and SQL Server.
- **Compliance & Controls:** Engineered a same-person verification guard enforcing segregation of duties and an immutable audit hash chain capturing 8 event types for tamper-evident, compliance-grade tracking.
- **AI Extraction & Quality:** Integrated AI-based invoice extraction with per-field confidence scoring and an AI Quality Dashboard (accuracy and P95 latency monitoring), supported by Hangfire background jobs for extraction and metrics.

Software Engineer, Cinnova Technologies

JULY 2022 - NOV 2025

Client: Nextek

Technologies: ASP.NET Core, ASP.NET MVC, Angular, JavaScript, SQL Server

- Contributed to the development of several business applications such as VirtualAdvisor, Generation Vault, and Campaign Builder.
- Focused primarily on Virtual Advisor, a powerful CRM system facilitating customer relationship management, sales processes, and internal workflows.
- Responsible for building, maintaining, and enhancing system modules to improve performance, scalability, and user experience.

Client: Wright Brothers Global Gas

Technologies: ASP.NET Core, Angular, JavaScript, SQL Server

- Designed a personalized Gas Distribution Management System for optimized administration of cylinder inventory, customer orders, delivery tracking, and billing.
- Delivered automation solutions that reduced manual work and improved customer-service efficiency.

Software Engineer, Information Technology University

SEP 2020 - JUNE 2022

WASA Tubewell Automation

Technologies: ASP.NET MVC, C#, HTML, CSS, Bootstrap, JavaScript, SQL Server

- Created a web-based monitoring system to automate watering tasks and eliminate manual effort wherever possible.
- Built dashboards, real-time reports, and a 3D HMI view to visualize parameters such as voltage, current, frequency, and motor status.
- Integrated fault-detection notifications for events such as power failure or inactive tubewells.

WASA Filtration Plants Automation

Technologies: ASP.NET MVC, C#, HTML, CSS, Bootstrap, JavaScript, SQL Server

- Designed and implemented an automation interface to supervise chlorine concentration, tank status, and water flow in filtration plants.
- Constructed real-time reporting, 3D visualization, and alarm modules for productivity tracking and troubleshooting.

EDUCATION

University of Engineering and Technology

2025 – 2027

MSc Artificial Intelligence — CGPA: 3.57

Research Topic: Failure Modes & Recovery in LLM Tool-Calling Pipelines

Punjab University College of Information and Technology

2016 - 2020

BS Software Engineering — CGPA: 2.72

Punjab College of Science

2014 - 2016

FSc Pre-Engineering — Marks: 83%

WAPDA Girls High School

2012 - 2014

Matriculation (Science) — Marks: 86%

ACADEMIC & DEVELOPMENT PROJECTS

Final Year Project: Intelligent Home Automation System

Technology: Machine Learning, C#, Android

Designed and developed an AI-powered Home Automation System consisting of two key modules:

- **Smart Door Module:** Implemented facial recognition using a 3D Convolutional Neural Network (3D-CNN) to authenticate and unlock doors for registered users.
- **Smart Appliances Module:** Developed an Android app and C# web interface for remote control of home electronic devices, integrating hardware with the software system for seamless automation.
- **Team & Entrepreneurship:** Pitched the project at Plan9 Incubation Center and secured incubation at Nspire; conducted user surveys, cost estimation, and market feasibility analysis, gaining hands-on exposure to the product-development lifecycle and startup ecosystem.

INTERPERSONAL SKILLS

- Self-Motivated
- Quick Learner
- Open to learning and exploring new technologies
- Leadership

ACHIEVEMENTS

- 81st Percentile in NAT
- IEEE certificate in a coding competition at UET
- Selected as a Startup at Netsol Incubation Center (Nspire)